AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 12 · STUDENT LESSON

Logs traces and privacy

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Failure budgets and service objectives. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of logs traces and privacy. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of logs traces and privacy.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of logs traces and privacy

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Failure budgets and service objectives. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain logs traces and privacy using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Logs traces and privacy case. Separate observations, given facts and assumptions before choosing an explanation.

Logs support diagnosis, but excessive recording can expose private content. **Trace ID**s link stages without storing complete sensitive **input**s.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Logs traces and privacy. Connect each labelled input, deciding step and result.

For this policy log **trace ID**, stage, status, duration and **error category**; do not log passcodes, student text or teacher answers. Restrict access and declare retention.

Logs traces and privacy — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A failure record includes trace R7, status=denied and the complete student passcode.	For this policy log trace ID, stage, status, duration and error category; do not log passcodes, student text or teacher answers. Restrict access and declare retention.	Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive error category.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Logs traces and privacy. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

A failure record includes trace R7, status=denied and the complete student passcode. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: A developer logs the full learner question.

Predicted result

Prediction: That additional field violates this logging policy.

Why the result follows

Reason: Debugging convenience does not authorise storing more personal information.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Logs traces and privacy to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Trace ID	An identifier connecting events within an operation
Minimisation	Keeping only data needed for the declared purpose
Error category	A non-sensitive label for a failure class

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Logs traces and privacy, predict a step, then compare with its explanation.

A failure record includes trace R7, status=denied and the complete student passcode.

Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

Worked explanation · Logs traces and privacy

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Logs traces and privacy

Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17.

The facts we will use

Supplied facts

R17 is the request **trace ID**; C8 is the retrieved source ID. Allowed fields also include validation pass and duration 240 ms. Raw question text is excluded.

AI ACADEMY / AI-6/12

Logs traces and privacy

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Connect events

R17 links the request, retrieval and validation events.



2. Inspect duration

240 ms describes this request, not the service average.



3. Minimise fields

Keep allowed metadata; omit private question text and names.

CHANGE ONE THING

A developer logs the full learner question.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Connect events	R17 links the request, retrieval and validation events.
2. Inspect duration	240 ms describes this request, not the service average.
3. Minimise fields	Keep allowed metadata; omit private question text and names.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Logs record events; traces connect related events. Useful debugging fields should remain within the declared privacy policy.

What this example does not establish

Keep the stated limit

Trace identifiers can be linkable too; specify their access, retention and deletion rules.

Keep the stated limit

Read the labelled picture: Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?

Keep the stated limit

Change one thing: A developer logs the full learner question. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Logs traces and privacy claim and explain a repair using the supplied evidence.

A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Logs traces and privacy with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Logs traces and privacy”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Logs traces and privacy. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Logs traces and privacy. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

A failure record includes trace R7, status=denied and the complete student passcode. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Logs traces and privacy. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

Add a labelled logs traces and privacy **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of logs traces and privacy.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

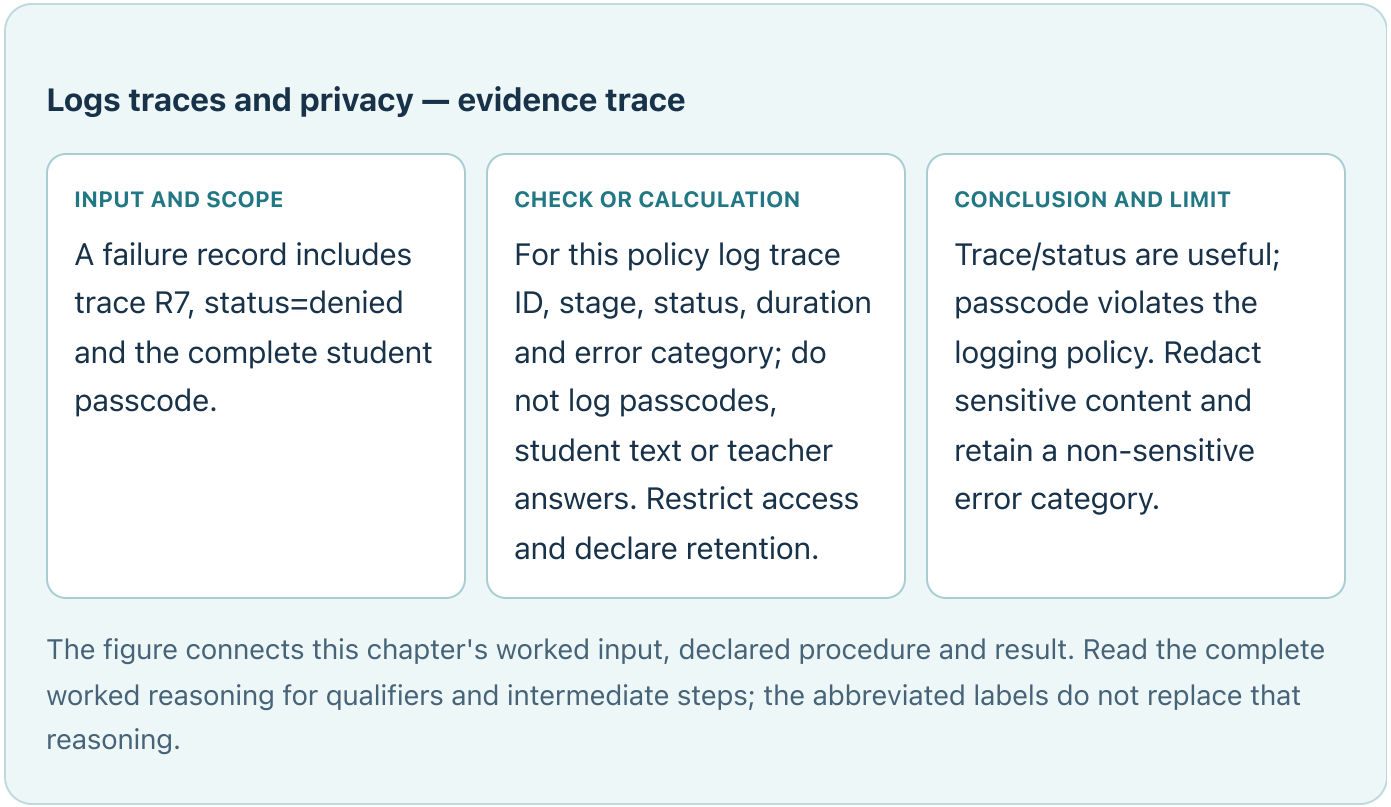
Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

Study a complete case

> Revisit the complete Logs traces and privacy case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Logs contain user essay, teacher guide, stage=lookup, duration=12ms and trace T8. Select allowed fields.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Logs contain user essay, teacher guide, stage=lookup, duration=12ms and trace T8. Select allowed fields.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Logs traces and privacy, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17.

By the end, I can	How I will show it
Explain the mechanism and terminology of logs traces and privacy	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Logs traces and privacy without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of logs traces and privacy

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Keep allowed metadata; omit private question text and names.

1. Inspect duration

2. Connect events

3. Minimise fields

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Minimise fields Keep allowed metadata; omit private question text and names. Logs record events; traces connect related events.

Useful debugging fields should remain within the declared privacy policy.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A developer logs the full learner question. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. That additional field violates this logging policy.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That additional field violates this logging policy. Debugging convenience does not authorise storing more personal information.

Repair a missing building block

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the **mechanism** and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Review Integration tests and fixtures (ai-6/9). Goal: Explain the **mechanism** and terminology of integration tests and fixtures. Short practice: Follow request N through the fixture and adapter.

Which source and item should appear in the reply, and what precisely fails when the adapter returns M?. Return to this week and explain what you changed.

Review Least privilege and tool scopes (ai-6/6). Goal: Explain the **mechanism** and terminology of least privilege and tool scopes. Short practice: Classify read S2, write S2 and read S5 as permitted or denied.

Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.. Return to this week and explain what you changed.

Keep your completed work

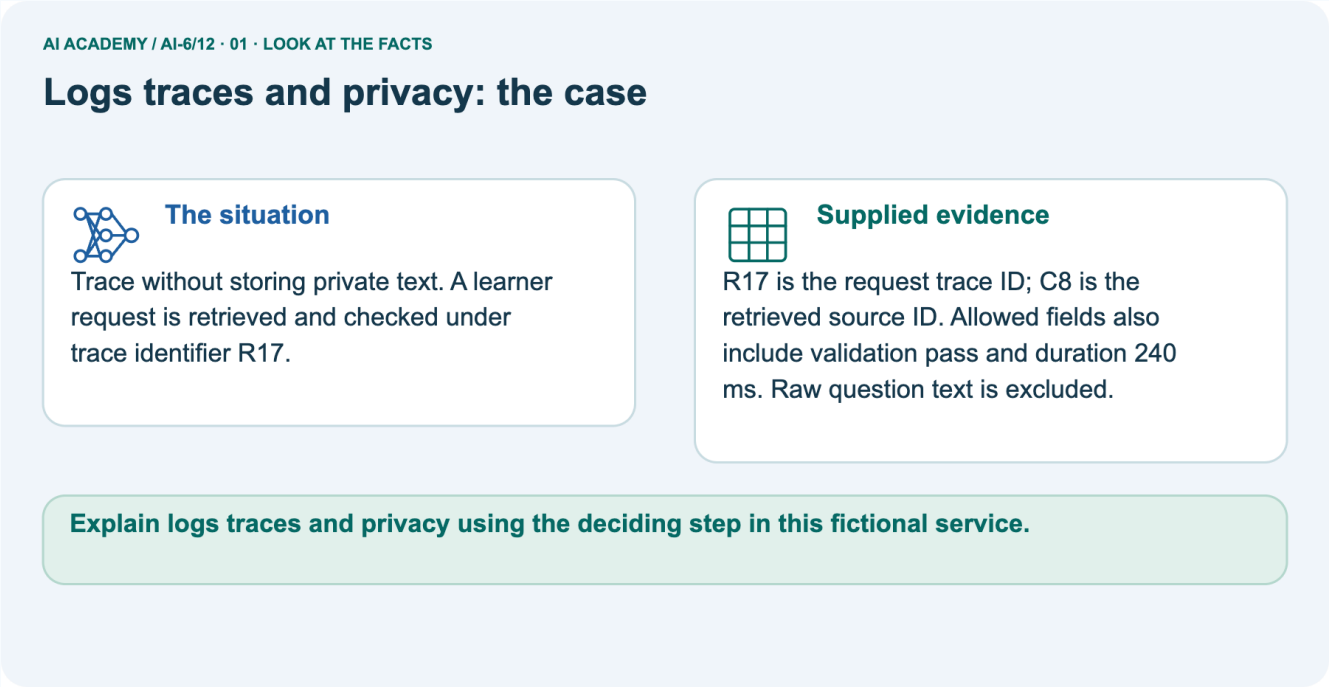
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Logs traces and privacy graphic using its labels, deciding evidence and relevant limitation.

01. Logs traces and privacy: the case



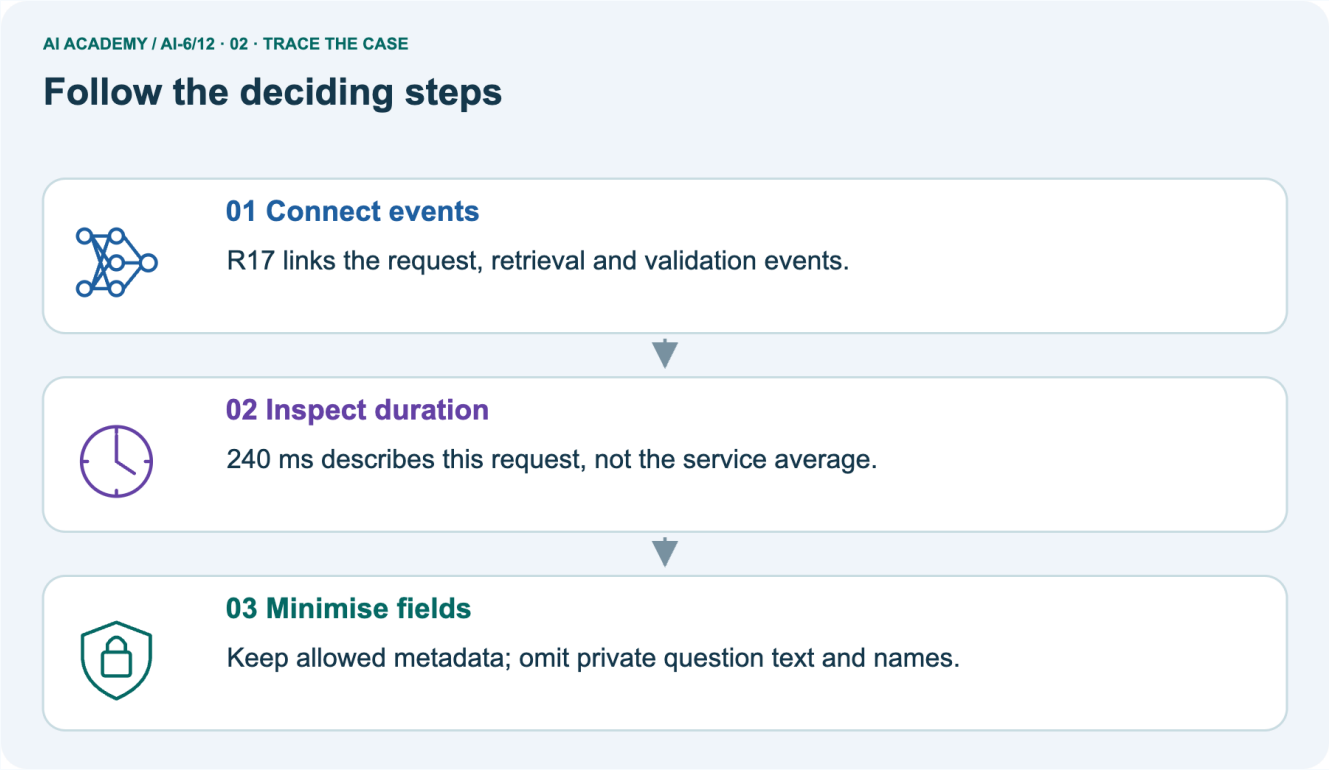
Follow the labelled picture with the matching explanation. Open the vector diagram

Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17. Supplied facts: R17 is the request **trace ID**; C8 is the retrieved source ID.

Allowed fields also include validation pass and duration 240 ms. Raw question text is excluded.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

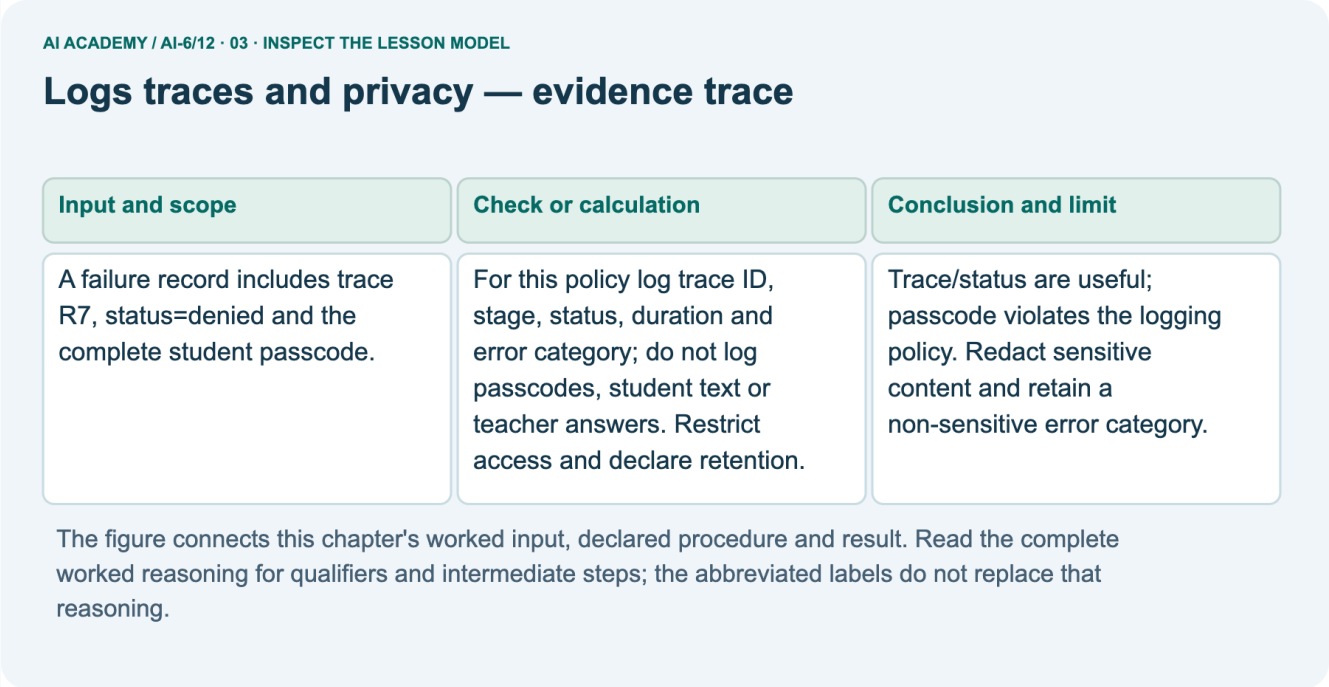


Follow the labelled picture with the matching explanation. Open the vector diagram

Connect events: R17 links the request, retrieval and validation events. → Inspect duration: 240 ms describes this request, not the service average. → Minimise fields: Keep allowed metadata; omit private question text and names.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Logs traces and privacy — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

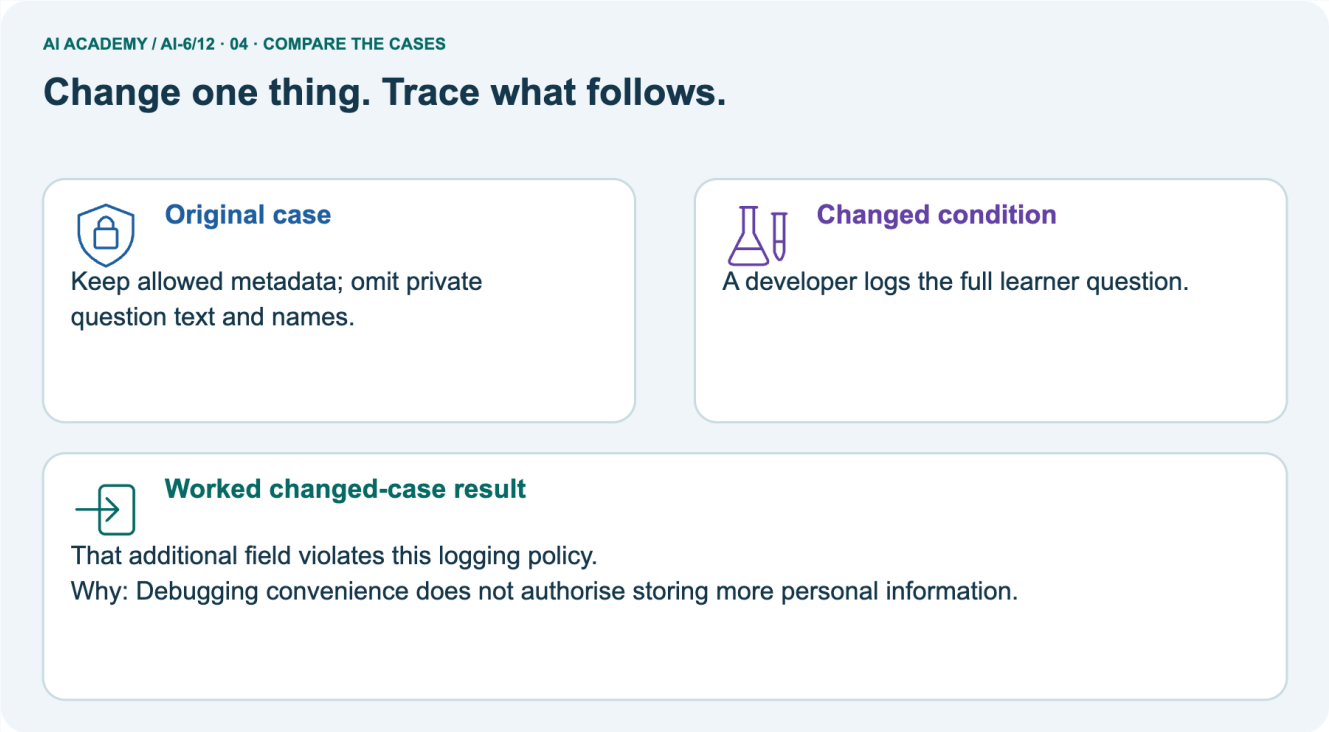
Logs traces and privacy — **evidence** trace.

A failure record includes trace R7, status=denied and the complete student passcode. | For this policy log **trace ID**, stage, status, duration and **error category**; do not log passcodes, student text or teacher answers.

Restrict access and declare retention. | Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



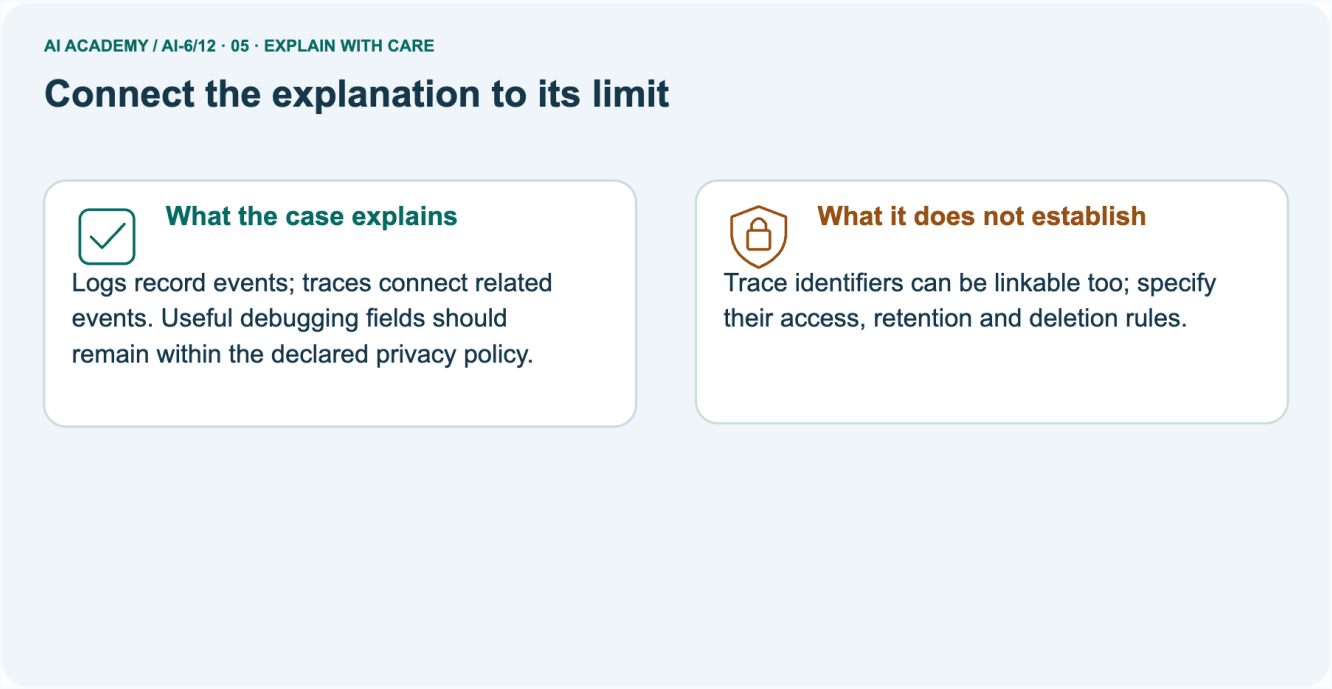
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep allowed metadata; omit private question text and names. Changed condition: A developer logs the full learner question. Expected result: That additional field violates this logging policy.

Reason: Debugging convenience does not authorise storing more personal information.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit

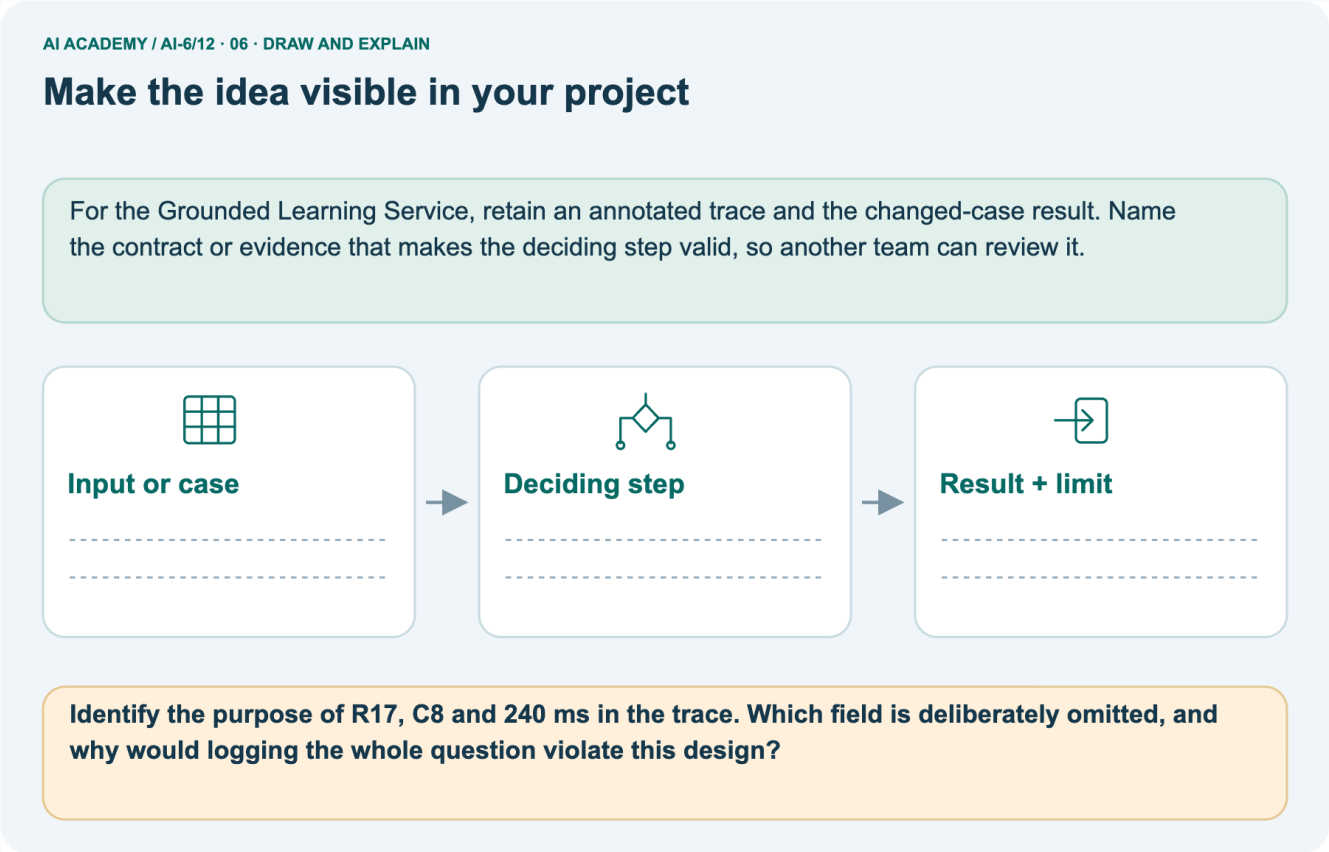


Follow the labelled picture with the matching explanation. Open the vector diagram

Logs record events; traces connect related events. Useful debugging fields should remain within the declared privacy policy. Limit: **Trace id**entifiers can be linkable too; specify their access, retention and deletion rules.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.